

Phase 3 one page summary

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What we have done

Reproduced the InstaSHAP tabular pipeline in a modular repository.

Built a focused Phase 3 extension around a real limitation: unrealistic transformed-space zero masking.

Implemented ``empirical_background`` masking so hidden feature groups come from real transformed training rows.

Evaluated the improvement on Covertypes with predictive, explanation, coalition, and runtime metrics.

Added an Adult Income masking diagnostic to show the masking improvement more clearly on a category-heavy dataset.

Current Covertypes result

Accuracy: zero 0.6843 vs bg 0.6774

Explanation MAE: zero 0.3591 vs bg 0.3795

Spearman: zero 0.5650 vs bg 0.5835

Coalition MSE: zero 0.2021 vs bg 0.2016

Interpretation: the new masking idea is valid, but the current end-to-end Covertypes result is mixed rather than a full win.

New Adult diagnostic result

Hidden categorical validity: zero 0.0000 vs bg 1.0000

Hidden categorical invalid rate: zero 1.0000 vs bg 0.0000